NS-RAD: The Full Project Report

From the 19-year climate-forced river record, through a campaign-year climatology, to a Green's-function sensitivity analysis of the planned FORTE field campaign

WHO THIS IS FOR: anyone who wants the complete story without prior familiarity with NS-RAD or the FORTE field campaign. No background is assumed -- every technical term is explained the first time it is used.

WHAT THIS REPORT COVERS, IN ORDER:

Part 1 The 2005-2023 long-term run: every forcing input, the model configuration, and 19 years of simulated river behavior
Part 2 How a campaign-specific climatology was built from that record
Part 3 The synthetic field-campaign sampling (the OSSE)
Part 4 Green's-function sensitivity -- boundary AND along-estuary
Part 5 Which measurements are actually most valuable, and why
Part 6 Conclusions

Built from: tools/build_interannual_*.py, run_interannual_full.sh, build_forte_campaign_forcing.py, run_forte_campaign.sh, build_osse_observations_forte.py, build_green_perturbations_forte.py, build_green_functions_forte.py, rank_observation_value_forte.py -- each documented in full in its own module docstring.

Executive summary

This report has two halves. Part 1 characterizes 19 years (2005-2023) of simulated river behavior on Alaska's North Slope, built from real, multi-source climate and ocean forcing. Part 2 onward asks a focused, practical question built on top of that record: which measurement types planned for the FORTE Expedition field campaign will actually teach us the most, and are there blind spots in how the campaign is timed or placed?

Part 1 in brief:

Three North Slope rivers (Colville, Kuparuk, Sagavanirktok) were simulated continuously for 19 years under real, independently-sourced daily-to-hourly forcing (discharge, meteorology, marine chemistry, tides, storm surge -- see Part 1's forcing-inventory table). All three show a significant INCREASING trend in discharge and DOC export, mirrored by a significant DECREASING trend in DIC/alkalinity -- a coherent dilution signal. Freshet timing, ice duration, and most reaction rates show no significant trend over this 19-year window (Part 1's findings page).

Parts 2-5 result, ranked:

1. Grab sampling -- score 12.81 (out of a maximum of 25)
2. Airborne (AVIRIS-3+drone) -- score 5.47 (out of a maximum of 25)
3. PACE (notional) -- score 2.97 (out of a maximum of 25)
4. Mooring (offshore) -- score 2.94 (out of a maximum of 25)
5. Mooring (inshore) -- score 1.69 (out of a maximum of 25)
6. SWOT (notional) -- score 1.00 (out of a maximum of 25)
7. Drifters -- score 0.97 (out of a maximum of 25)

The one finding that matters most:

Grab sampling should be nearly worthless in FORTE's June-September window -- the ocean connection that drove its value in a full-year analysis is severed by the spring freshet before the campaign starts. But grab sampling is ALSO sensitive to the river's own upstream chemistry and to what is happening ALONG the channel in between -- testing both (Parts 4-5) flips the conclusion: grab sampling comes out on top, not the bottom.

Part 1: What is NS-RAD?

NS-RAD (North Slope River-Aquatic-Delta Model) is a one-dimensional reactive-transport simulation of Arctic river-estuary systems, built on the C-GEM modeling engine (Volta et al. 2014). It represents each river as a channel running from its tidal mouth upstream, resolving salinity, temperature, dissolved oxygen, carbonate chemistry (DIC, alkalinity, pH), nutrients (nitrate, ammonium, phosphate, silica), organic carbon, suspended sediment, phytoplankton biomass, and a prognostic sea-ice cover, all advected, dispersed, and reacting together through time.

Three North Slope rivers are configured and runnable: the Colville, Kuparuk, and Sagavanirktok. A fourth (Canning) has no observed estuary length yet and is excluded. Each river's geometry (width, depth, estuary length), tidal constituents, and boundary chemistry are observation-based and site-specific; the reaction network and numerics are shared across all three.

Like any model of a real place, NS-RAD needs to be told what is happening at its EDGES -- the BOUNDARY CONDITIONS -- since it cannot simulate the whole ocean or the whole river basin:

UPSTREAM boundary ("cub"): what is flowing INTO the river from further inland -- discharge, and the chemistry of that water.

DOWNSTREAM/marine boundary ("clb"): what the OCEAN looks like at the river mouth -- salinity, temperature, carbon, nutrients, and the tide/storm-surge water-level signal.

Everything in this report traces back to these boundaries: Part 1 describes how they (and the meteorological forcing above the channel) are built for a genuine 19-year simulation; Parts 2-5 ask how well planned field measurements could pin them down.

Part 1: Why a 19-year run, and why 2005-2023?

Every NS-RAD run before this one used a single repeating "typical year" of forcing (originally 2022). That is sufficient for validating the model's physics against observations, but cannot answer a different kind of question: how much does river behavior actually VARY from year to year, and is there a detectable multi-year TREND underneath that variability? Answering that needs a genuinely multi-year run, with real year-to-year forcing, not a repeated climatology.

WHY 2005-2023, NOT A LONGER WINDOW: this is the shortest common window across every independent forcing source this run needs, checked directly against each source's own archive rather than assumed from documentation (see Part 1's forcing-inventory table, next page, for the full per-source coverage). The binding constraint is wind and air temperature: the nearest continuous meteorological station did not exist before 2005. Three alternatives were considered and declined: switching the whole meteorological stack to a reanalysis product (would extend the window to 1995, but adds a credential requirement this project has otherwise avoided), accepting a different window per source (declined in favor of one common window every forcing category shares), or restricting discharge -- which has real data back to 1980 -- to match everything else anyway. The last option is effectively what was chosen: every source is sliced down to the shortest common window, 2005-2023, 19 years, so that every single simulated year in this run is driven by entirely real, independently sourced forcing in every category at once -- not a mix of real-this-year and climatological-that-year.

Part 1: Every forcing input to the 2005-2023 run

Forcing	Source	Coverage / resolution	Notes
River discharge + riverine organic carbon (TOC)	PWBM (Pan-Arctic Water Balance Model)	1980-2023, daily	Whole-basin simulated flux + DOC concentration per river
Wind speed, air temperature	NDBC buoy PRDA2 (Prudhoe Bay)	2005-2023, hourly	THE binding constraint on the window -- station began 2005
Relative humidity	Deadhorse Airport (NOAA ISD)	2005-2023, hourly	Co-located with PRDA2
Solar radiation	NOAA GML Barrow (Utqiagvik) Atmospheric Baseline Obs.	2005-2023, 1-min (aggregated hourly)	Real pyranometer observation, not a reanalysis -- see note below
Storm surge (non-tidal sea-level residual)	NOAA CO-OPS 9497645, Prudhoe Bay	2005-2023, daily mean	Regional proxy for all 3 rivers (only station with continuous water level)
Marine (clb) boundary -- S, DIC, ALK, NO3, NH4, PO4, dSi, O2, TOC, sea-T	ECCO-Darwin v5 (ocean reanalysis)	2005-2023, monthly (interpolated daily)	Nearest wet grid cell to each river's mouth
River temperature (upstream boundary)	Regional air->water regression (this project)	2005-2023, derived from air temperature	T_river = max(0, 1.432*(T_air,10-day + 4.0)); fitted on Kuparuk, applied to all 3
Tidal elevation	NOAA CO-OPS harmonic constituents, per river	Continuous (no data file needed)	Computed analytically from elapsed model time -- genuine multi-year spring-neap for free

Solar radiation's source changed twice before landing on Barrow: a satellite product doesn't reach this latitude before 2013, and a 3-hourly reanalysis (NARR) was tried but superseded by this real, finer, more reliable station record. Every other row (except river temperature, a regional regression, and tides, computed analytically) is a genuine, independently-sourced observation -- not a reanalysis or a model product standing in for one.

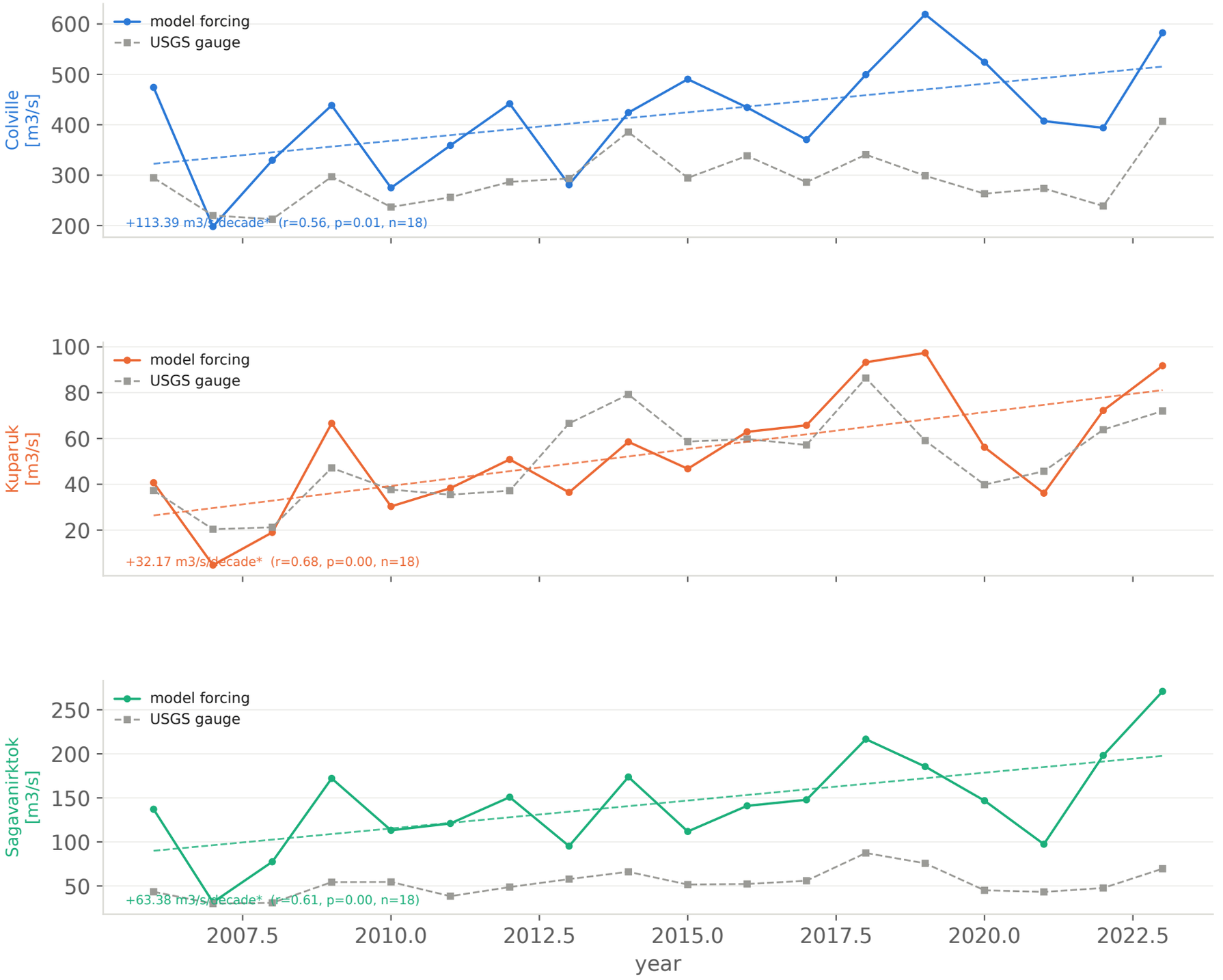
Part 1: Model configuration for this run

Setting	Value
Simulated period	6935 days (19 x 365) + 365-day warm-up = 7300 days total
Warm-up	First 365 days: hydrodynamics/transport only, biogeochemistry off
Grid spacing (DELXI)	100 m (all 3 sites)
Timestep (DELTI)	30 s (all 3 sites)
Output save cadence	Daily -- a 6-min cadence over 19 years would be tens of GB/site; daily is enough for annual budgets and seasonal shape
Estuary length (EL)	Colville 4150 m / Kuparuk 7156 m / Sagavanirktok 2170 m -- each river's own observed length, not a shared value
Grid points (M)	Colville 42 / Kuparuk 72 / Sagavanirktok 22
Ice model	Prognostic (freeze-up, conductive growth, bottom-fast grounding, surface melt, hydraulic breakup) -- on by default
Multichannel geometry	On -- braided-delta distributary width distinct from upstream prismatic per-thread width
Width/dispersion model	"Flare" width convergence + Seo & Cheong (1998) local shear dispersion -- see model documentation
Arctic BGC extension	Off (bit-identical to the shipped reaction network)

This is the SAME reaction network, numerics, and geometry as every other NS-RAD configuration -- only the forcing is new. Every solute boundary except temperature and TOC is still a CONSTANT at the upstream end (no time-varying source exists yet for riverine nutrients/carbonate); the marine end is now genuinely time-varying for every species, via ECCO-Darwin v5.

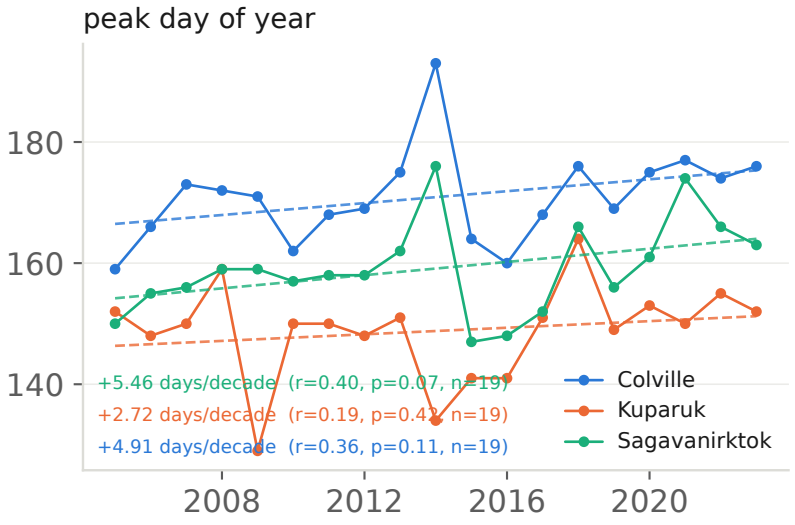
Part 1 results: annual discharge, 2006-2023

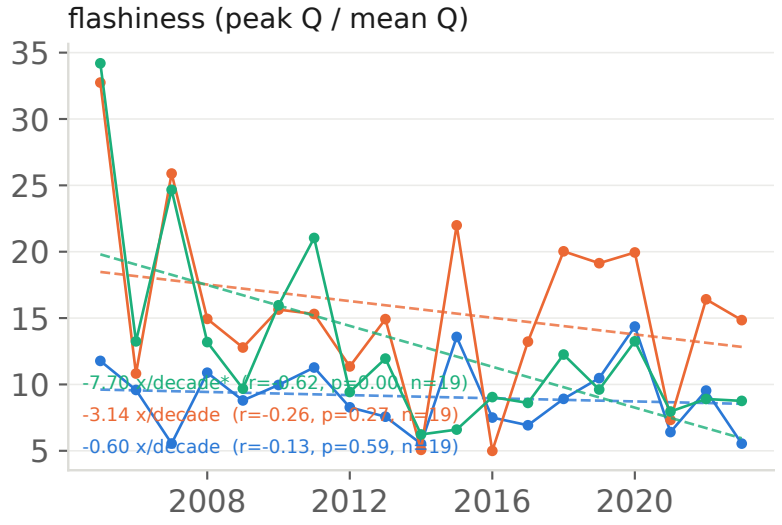
Model forcing (solid) vs. the real USGS gauge record (dashed), per river. Colville and Sagavanirktok are gauged well upstream of their deltas and are known to understate mouth discharge (documented model limitation, not new here); Kuparuk's gauge is near-tidewater and the best-constrained of the 3.

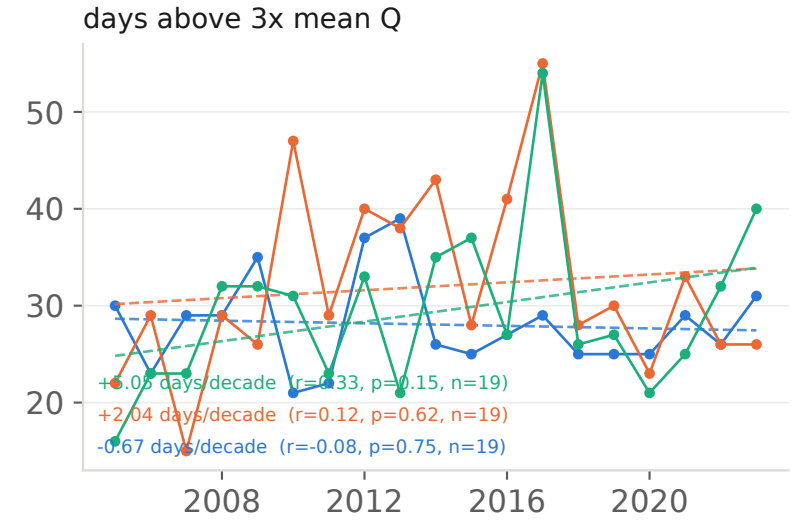


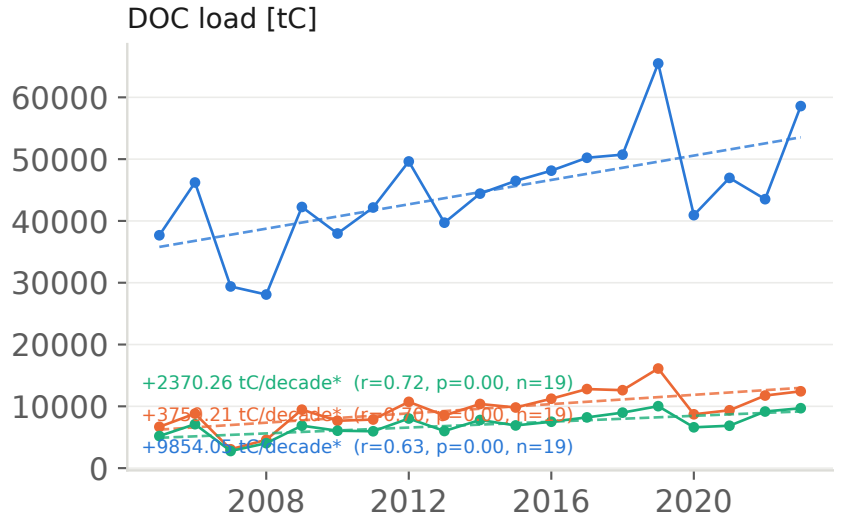
Part 1 results: spring-freshet characteristics, 2006-2023

Four freshet metrics per river per year, each with an OLS trend line (dashed, * = p<0.05). Duration and DOC load are the two genuinely diagnostic quantities here for land-to-ocean carbon delivery.



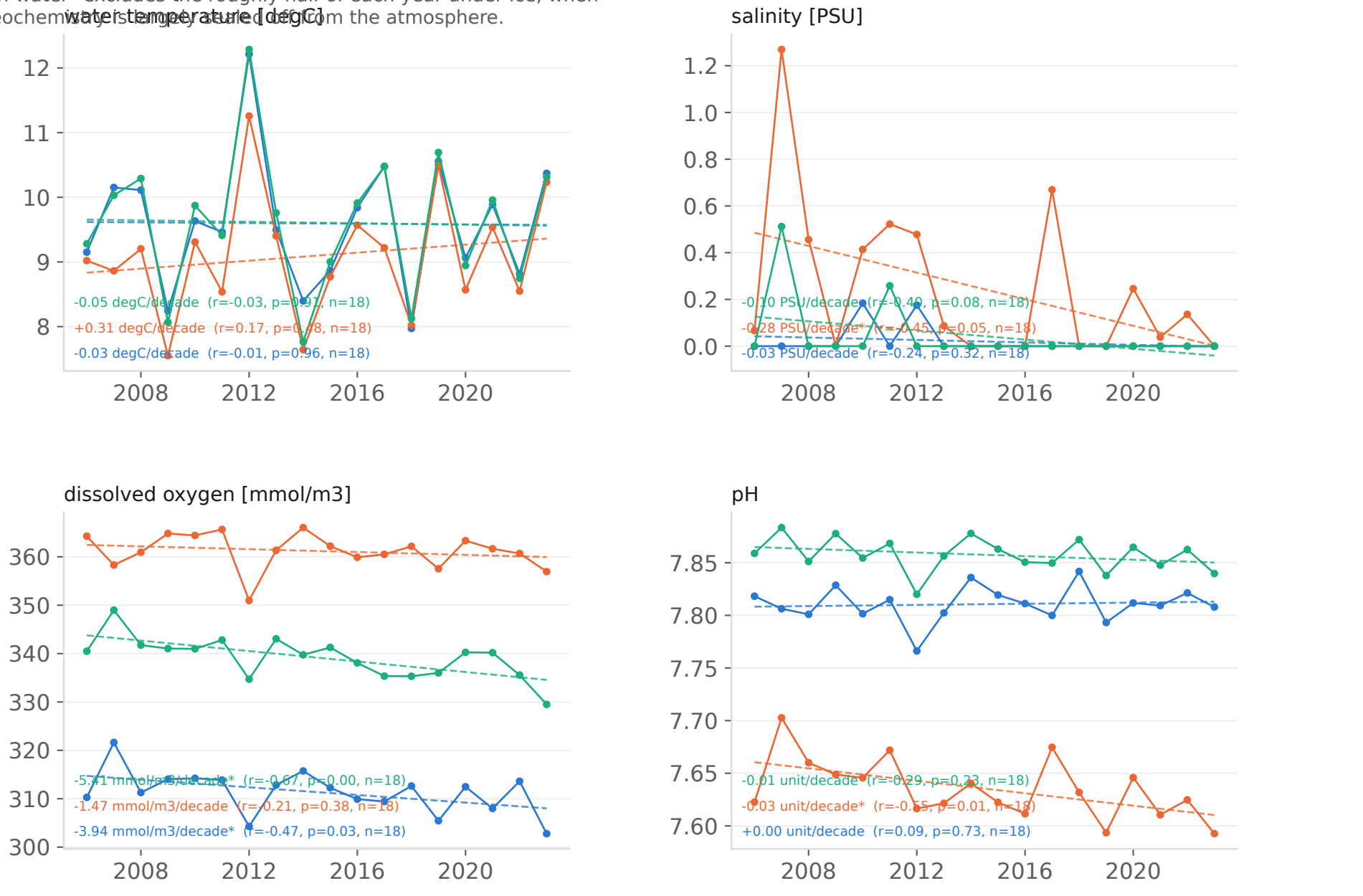






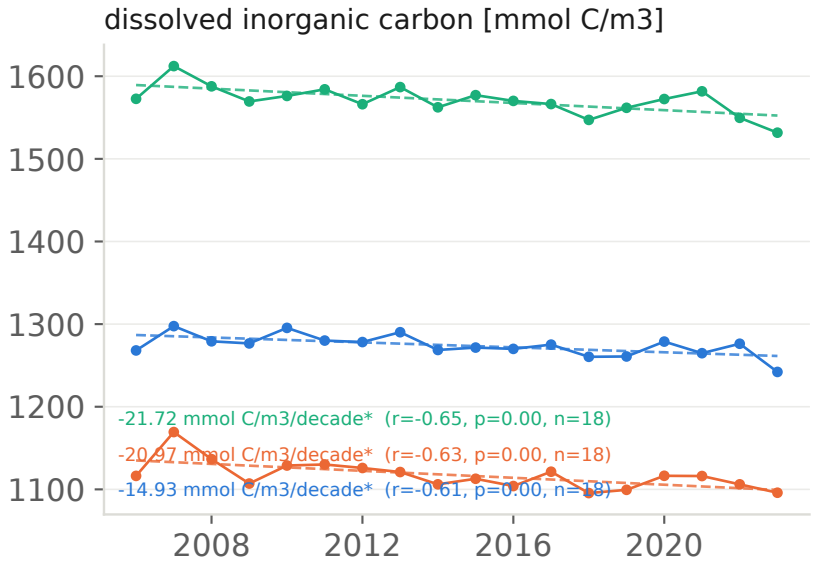
Part 1 results: major state variables, annual open-water means (1 of 2)

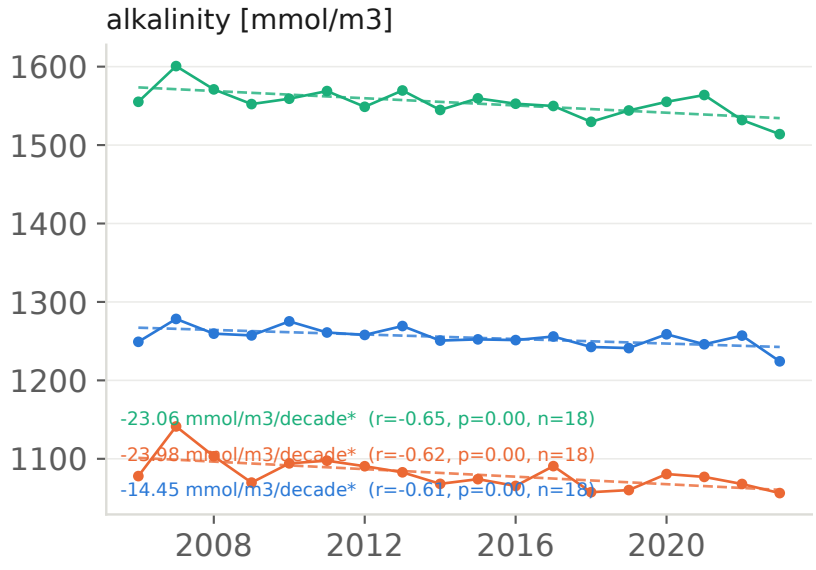
Spatial + temporal mean over open-water (ice-free) cells, per calendar year.
"Open-water" excludes the roughly half of each year under ice, when
biogeochemistry is largely decoupled from the atmosphere.

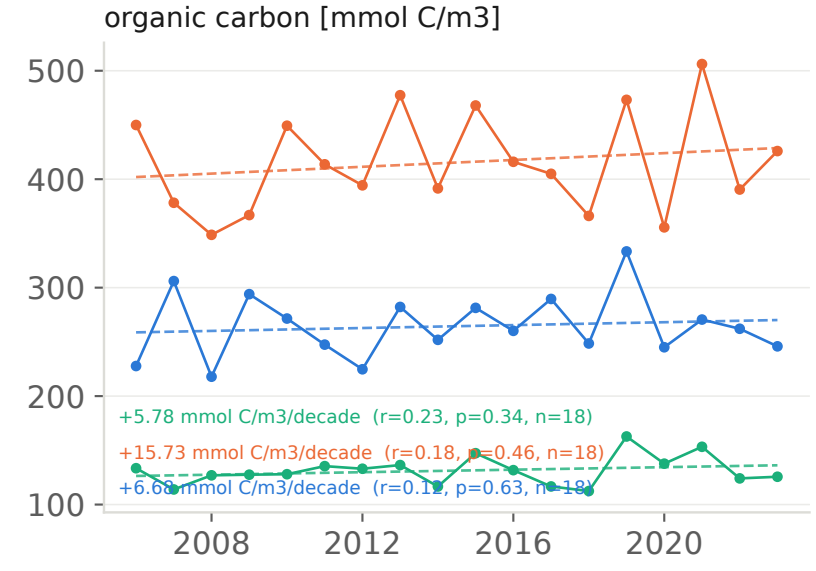


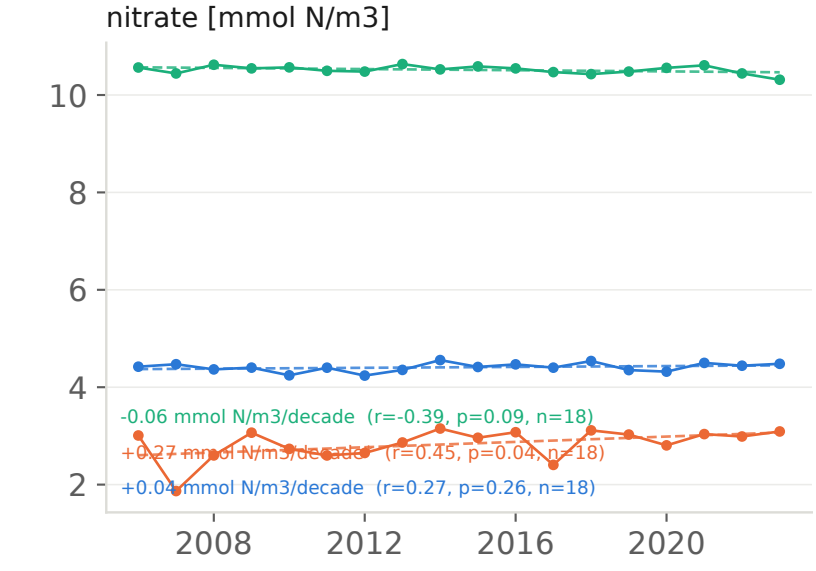
Part 1 results: major state variables, annual open-water means (2 of 2)

Same method as the previous page, for the carbon/nutrient system.



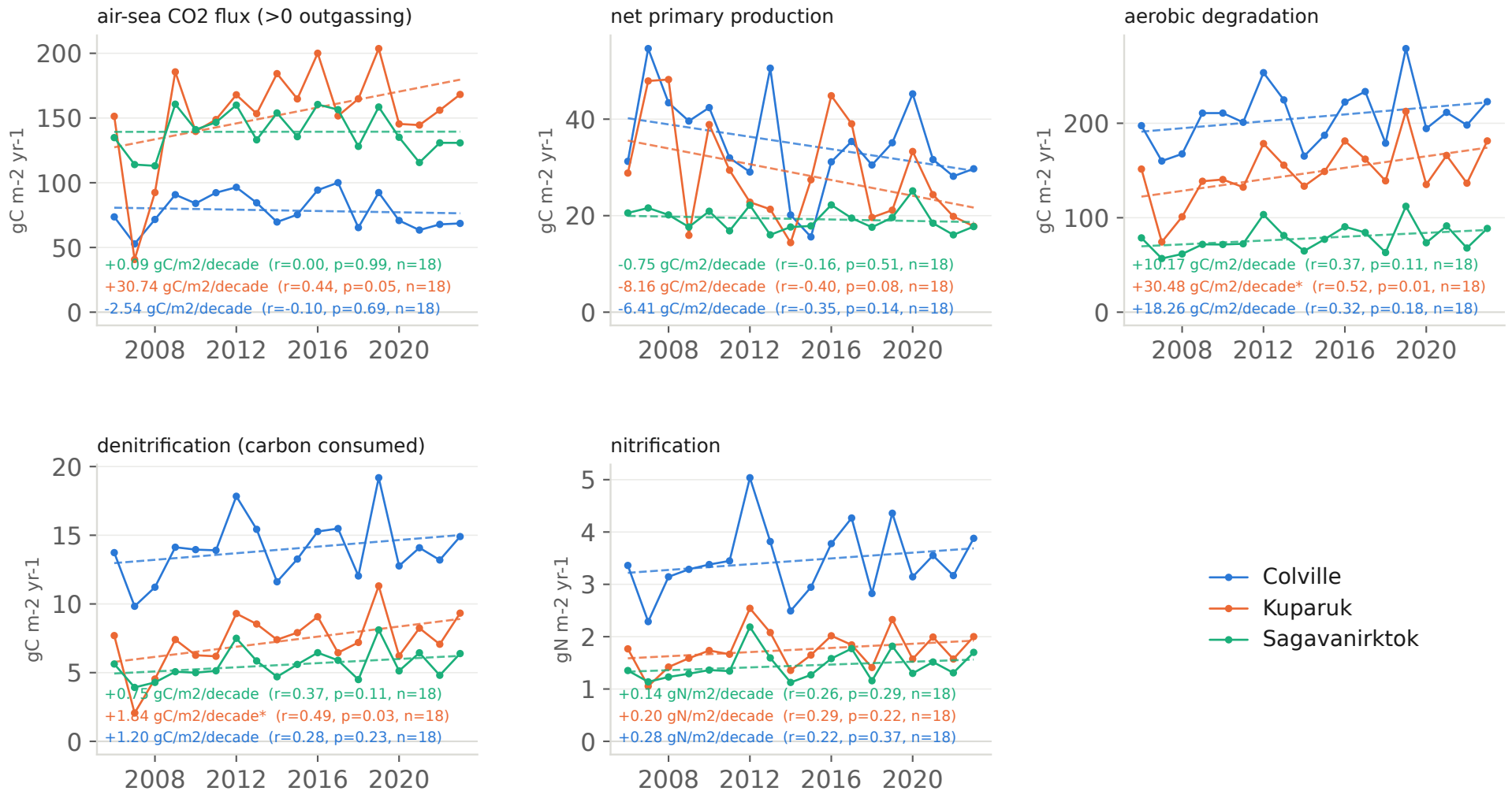






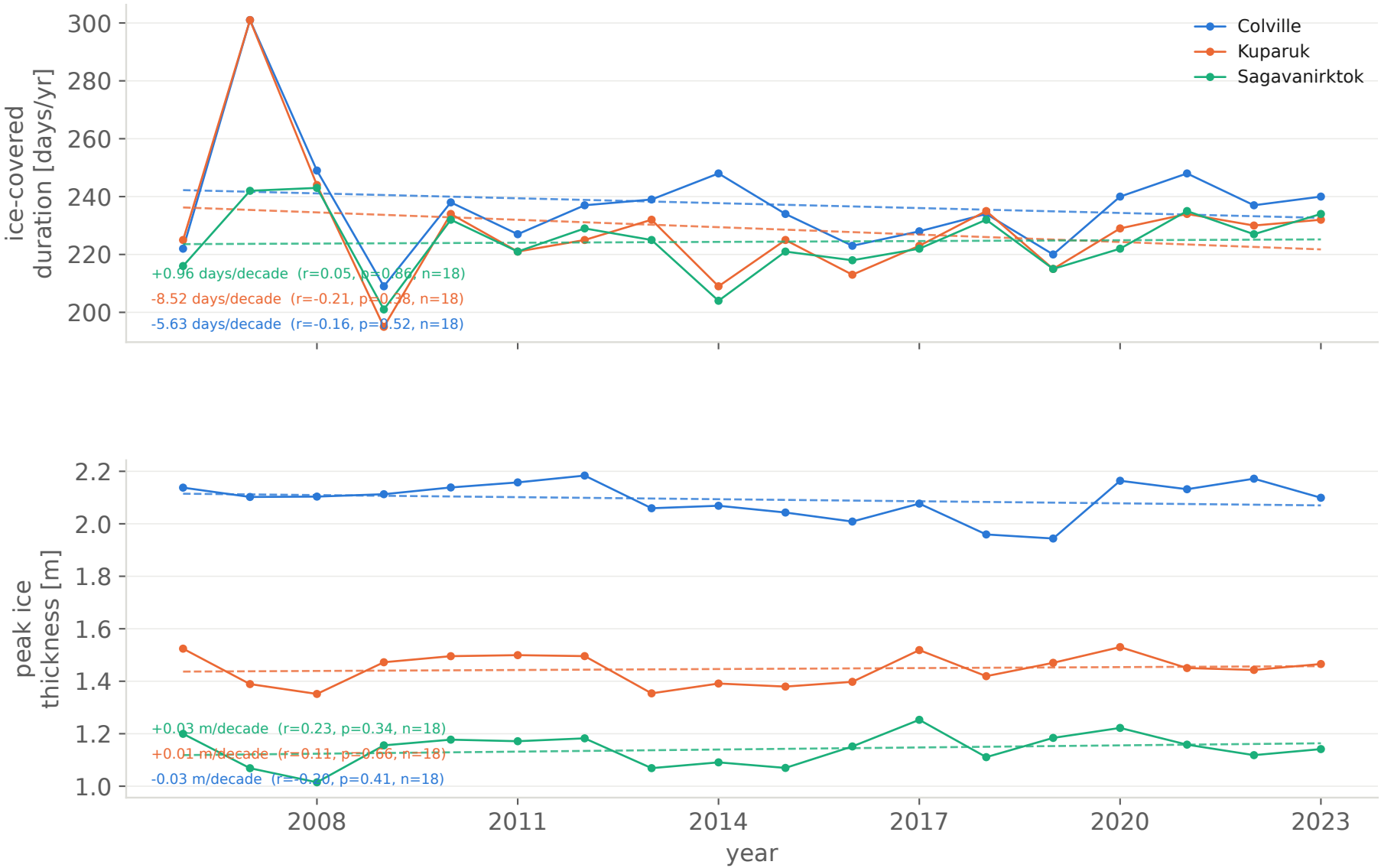
Part 1 results: area-normalized annual process-rate budgets

Each rate depth-integrated, width-weighted, and time-integrated over one calendar year -- same method as the 44-year record's own report, applied here to the 19-year full-forcing one.



Part 1 results: ice-cover variability, 2006-2023

Ice-covered duration (domain-mean ice fraction > 0.1) and peak thickness,
per calendar year -- the prognostic ice model's response to 19 real winters.



Part 1 summary: what 19 years of real forcing actually shows

1. A CLEAR, COHERENT SIGNAL IN DISCHARGE AND CARBON EXPORT. All three rivers show a statistically significant (p<=0.01) INCREASE in annual discharge (+32 to +113 m3/s/decade) and in annual DOC load (p=0.00 for all three) over 2006-2023. DIC and alkalinity show the mirror-image signal -- a significant DECREASE at all three rivers (p=0.00) -- consistent with a simple, physically coherent story: more freshwater runoff dilutes the river's background carbonate chemistry while carrying more organic carbon with it. This is a genuine multi-river, multi-variable pattern, not an isolated single-panel trend.

2. MOST OTHER QUANTITIES SHOW NO SIGNIFICANT TREND OVER THIS WINDOW. Freshet timing, flashiness, and duration; ice-covered duration and peak thickness; salinity, temperature, pH; and most reaction rates (NPP, nitrification, and aerobic degradation/denitrification at 2 of 3 rivers) all show p>0.05 trend lines -- reported honestly as null results, not suppressed. 19 years is a short baseline against this coast's substantial year-to-year weather noise, and a quantity NOT trending is as informative here as one that is.

3. DISCHARGE FORCING TRACKS THE REAL GAUGE RECORD. Kuparuk (the best-constrained gauge) shows close agreement in both magnitude and year-to-year variability with the real trend; Colville and Sagavanirktok run systematically higher, consistent with their already-documented upstream-gauge limitation (both gauged well above their deltas) -- but all three still show the SAME significant increasing trend as their own gauge records, so the forcing is not manufacturing the signal.

4. THIS RECORD IS THE FOUNDATION FOR EVERYTHING THAT FOLLOWS. Parts 2-5 do not use this 19-year record directly -- they use a CLIMATOLOGY derived from it (Part 2), so the choices and data quality documented here propagate forward into the FORTE campaign analysis, not just into this Part's own figures.

Part 2: Why not run the real campaign dates directly?

The FORTE Expedition's real target is 2027 (and possibly 2028) -- years this model cannot simulate directly, since its forcing data only goes through 2023. Two further considerations shaped what to build instead:

NS-RAD HAS NO RESTART/CHECKPOINT CAPABILITY. Every run must simulate from day zero; there is no way to "resume" a long run partway through. Running an entirely fresh multi-year simulation for every possible future campaign year would be prohibitively expensive and would still not reach 2027.

A SINGLE ANALOG YEAR WAS CONSIDERED AND DECLINED. Picking one historical year (e.g. the driest, or the most typical) to stand in for "what might happen in 2027" is somewhat arbitrary and commits the analysis to that year's idiosyncrasies. Instead (project owner's explicit choice, 2026-10-19), a CLIMATOLOGY was built: every forcing input is averaged across all 19 real years, day-of-year by day-of-year (or hour-of-year for sub-daily forcings), into one smooth, representative "typical campaign season." This is reusable for any future campaign year without re-choosing an analog, and is exactly the kind of baseline real field-campaign data (especially a genuinely time-varying upstream boundary, which does not exist for any historical year, real or synthetic) is meant to eventually be compared against or substituted into.

This is a new, separate framework alongside the existing 19-year run (Part 1) and an earlier 3-analog-year OSSE exercise -- both of those stay exactly as they are, for their own existing purposes.

Part 2: How the climatology was built

For every forcing file in Part 1's inventory (discharge, meteorology, marine boundary chemistry, storm surge, river temperature), the climatology is computed directly from the already-validated 2005-2023 series:

DAILY forcings (discharge, TOC, marine chemistry, surge, river temp): reshape the 19 x 365-row series to (19, 365) and average across the 19 years, giving one 365-row "typical year" file.

HOURLY forcings (wind, air temperature, humidity, solar): the same recipe, reshaped to (19, 8760), giving one 8760-row file.

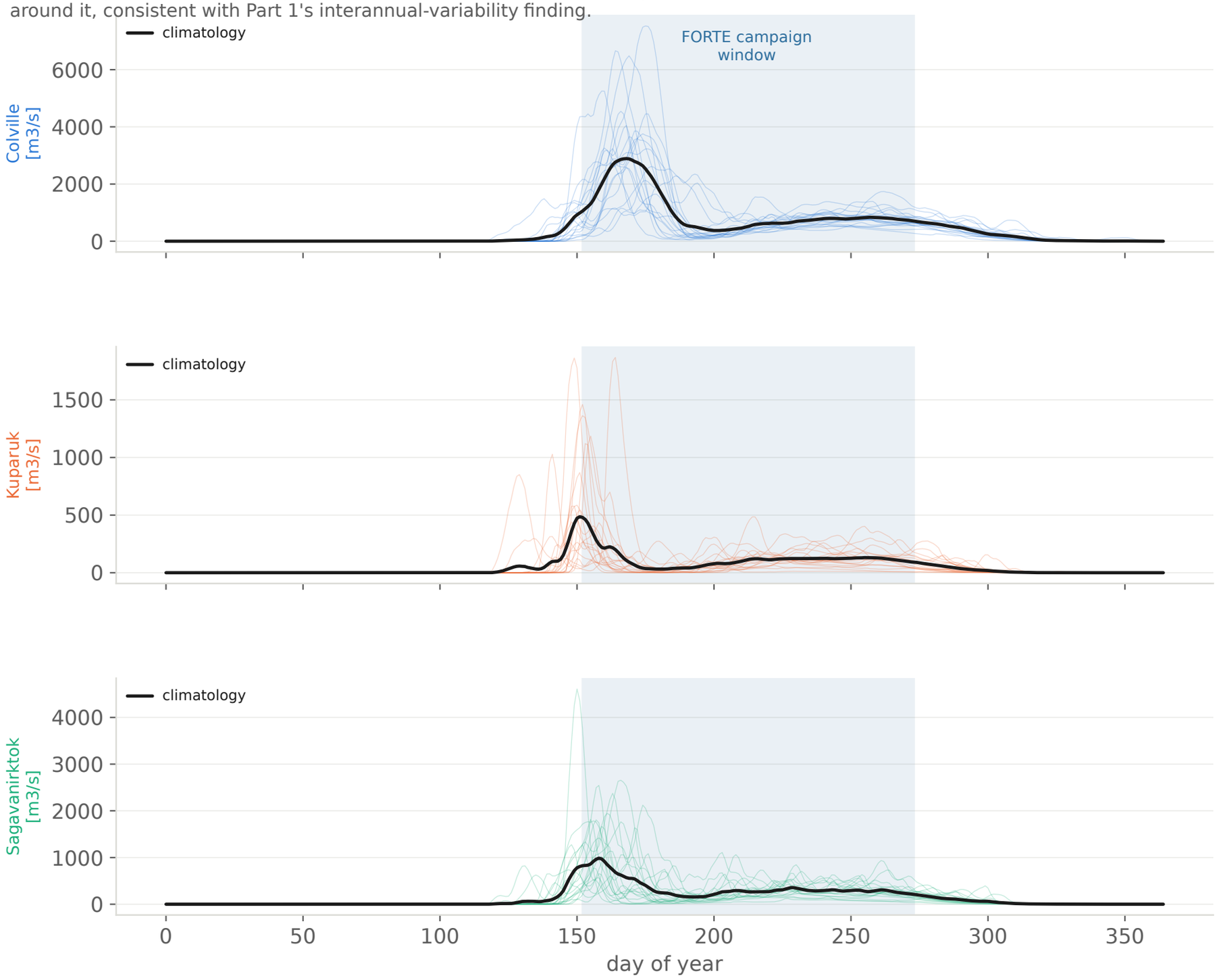
Verified directly before use, not assumed: every *_interannual_2005-2023_* .csv is exactly 19*365 (daily) or 19*8760 (hourly) rows, with no leap days (matching every other NS-RAD forcing file's calendar convention).

These climatology files feed a new site variant (sites/<river>_forte_campaign.py) pointing at them instead of the 19-year series -- no model-code changes needed, since repeatYear already wraps any 365-row/8760-row file exactly the way every single-year site always has.

Tides need no new file at all: the harmonic tide is computed analytically from elapsed model time, so it is already a genuine (if idealized, non-calendar-pinned) multi-year signal in every NS-RAD run.

Part 2: the climatology against the 19 real years it summarizes

Thin lines: each of the 19 real years' own discharge, day-of-year. Thick line: the climatology (their day-of-year mean) that actually drives the FORTE campaign run. The climatology tracks the central tendency faithfully; individual years' freshet timing and peak magnitude vary substantially around it, consistent with Part 1's interannual-variability finding.



Part 2: the FORTE campaign run and the planned field campaign

Phase	Real dates (2027)	Day-of-year	What happens	Detail
Phase 0	Late April (TBD)	--	Under-ice	NOT part of this analysis -- infeasible per the campaign plan (rivers frozen, negligible flow)
Phase 1a	Jun 1-5	152-156	Grab sampling	Helicopter-accessed water samples; DOC, nutrients, DIC, alkalinity, chlorophyll-a
Phase 1b	Jun 6-20	157-171	Grab sampling + ice drone	Continues grab sampling; separate ice-thickness drone survey (not analyzed here)
Phase 2	Jul 1-12	182-193	Airborne + moorings deployed	Aircraft-based optical imaging; fixed instruments go into the water
Phase 3	Jul 20-Aug 7	201-219	Airborne+drone + offshore moorings	Continues airborne imaging; moorings stay in through September

The run itself: climatological forcing from Part 2's method, MAXT_DAYS=638 (365-day warm-up + 273 days covering the full campaign season), 6-minute output cadence (needed for the Green's-function perturbation work in Part 4). Every comparison from here on uses the campaign's CALENDAR DATES (day-of-year) applied to this climatological run, not literal 2027.

Part 3: What is an OSSE, and why build one before the real campaign?

An OSSE -- Observing System Simulation Experiment -- is a standard technique used across climate and ocean science before an expensive field campaign happens. The idea:

1. Treat the model's own simulated state as a stand-in for "the truth" (here, the Part 2 climatological campaign-year run).

2. Simulate what each planned instrument would ACTUALLY measure if it sampled that "truth" -- at the right times, the right places, with the instrument's own realistic measurement error. Done here for all five of FORTE's measurement types (next page).

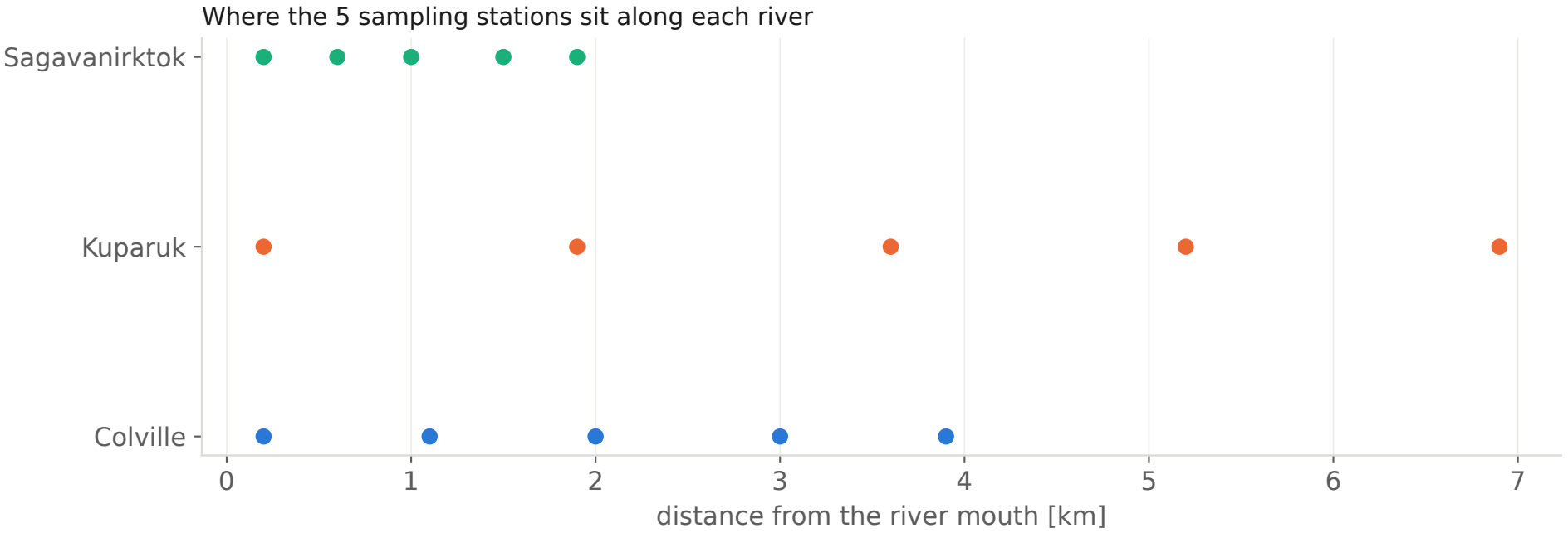
3. Ask: if an instrument had only ever seen these synthetic measurements, how much would it actually have learned? Answered in Part 4-5 via a Green's-function sensitivity analysis.

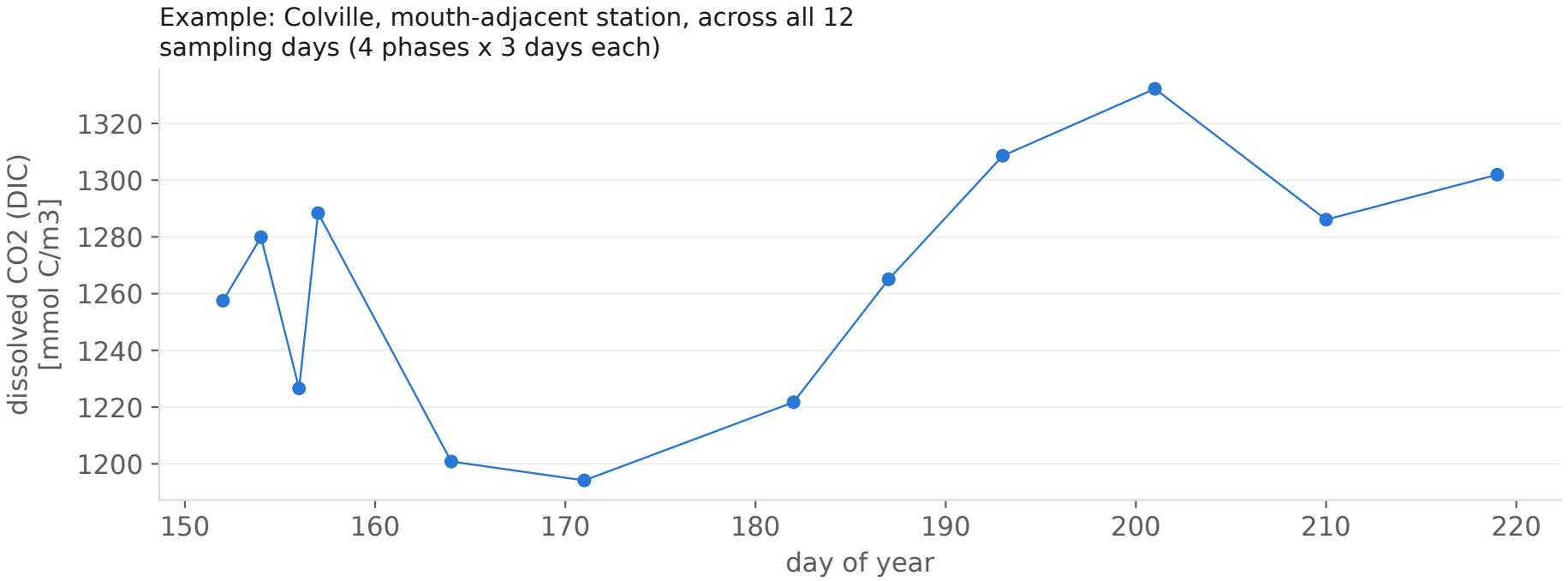
The payoff: problems with a campaign's DESIGN (not needing more funding or better instruments, but better TIMING or PLACEMENT) can be found and fixed before a single real measurement is taken.

Part 3: the five measurement types this report evaluates

Measurement type	Where	How often	What it measures	In plain terms
Grab sampling	5 stations/river (along the length of the estuary)	4 phases x 3 days/ phase (first/mid/last)	Dissolved organic carbon, nutrients, dissolved CO2/alkalinity, chlorophyll-a	Water samples collected by hand, analyzed in a lab
Moorings	1 fixed point inside the estuary + the ocean itself	Continuous, Jul 1 - Sep 30	Salinity, temperature, dissolved oxygen, current speed	Instruments left in place for months, recording continuously
Drifters	Follow the water (float with the current)	30 released across the season	Salinity, temperature, and how long it takes to drift out to sea	Small floating sensors ("microSWIFT") that travel with the current
Airborne optical	Every point along the estuary, in one pass	Every 8 hours, Jul-Aug only	Water color/clarity -- proxies for organic carbon, algae, sediment	Camera/sensor flown overhead on a plane or drone
Satellite (notional)	Wide river mouths only (SWOT); the plume (PACE)	SWOT ~every 3 weeks; PACE ~daily	Water-surface height (SWOT); water color (PACE)	Not a committed part of the real campaign -- included as a "what if"

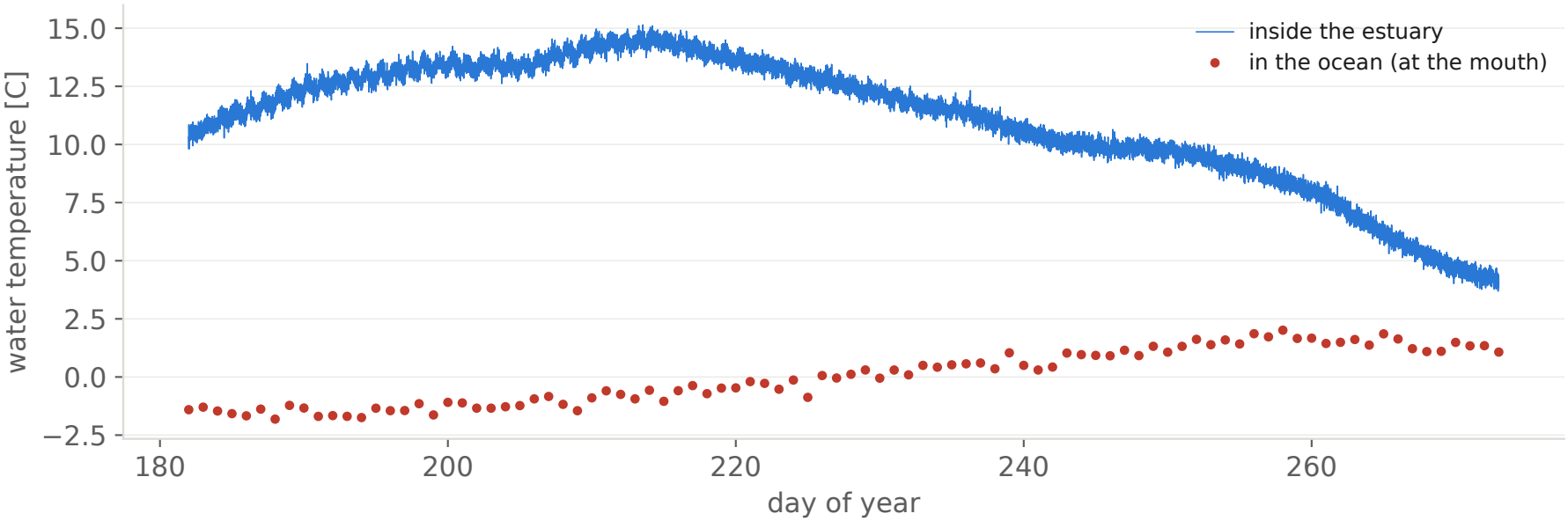
Part 3 results: what the synthetic grab samples actually look like



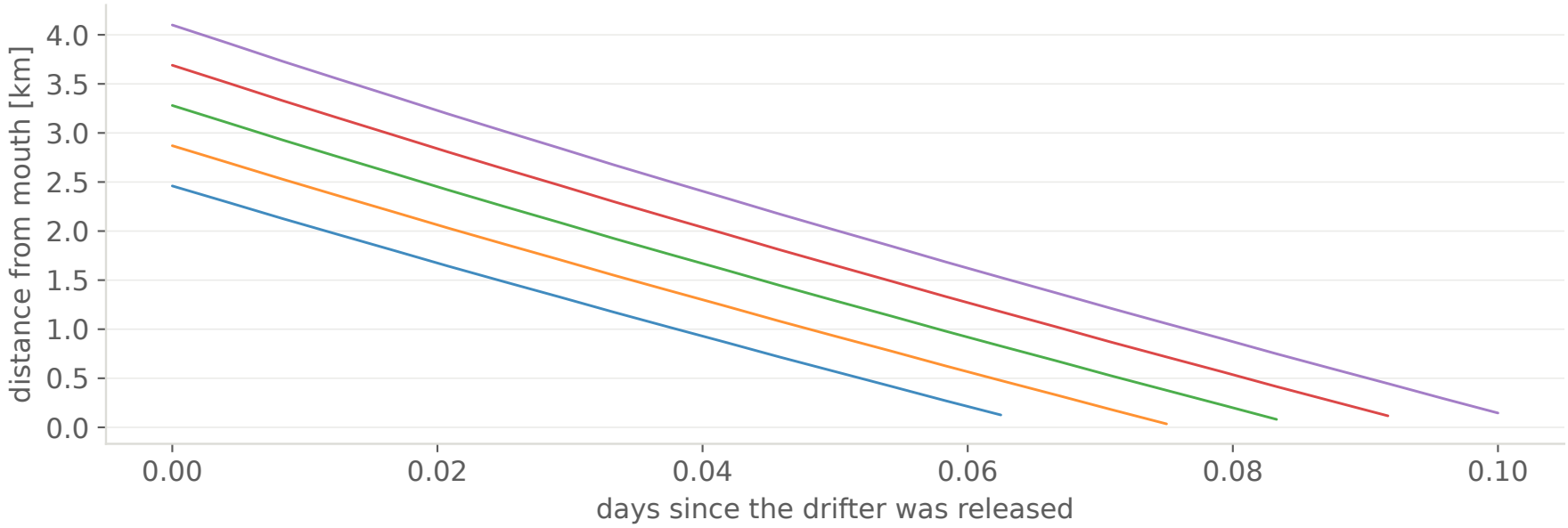


Part 3 results: moorings and drifters

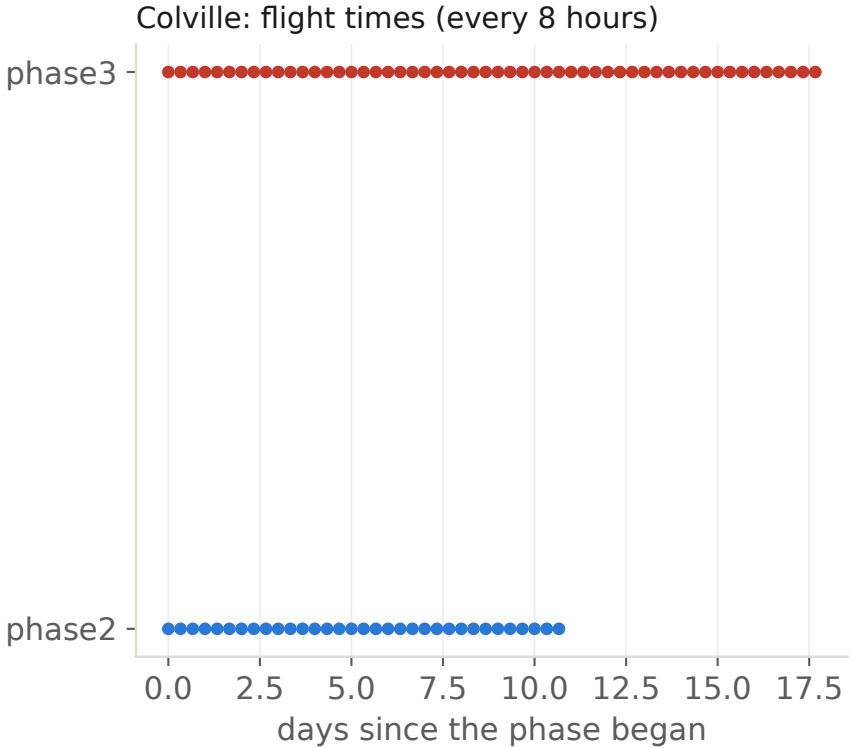
Colville: a mooring inside the estuary runs far warmer in summer than the ocean right outside it

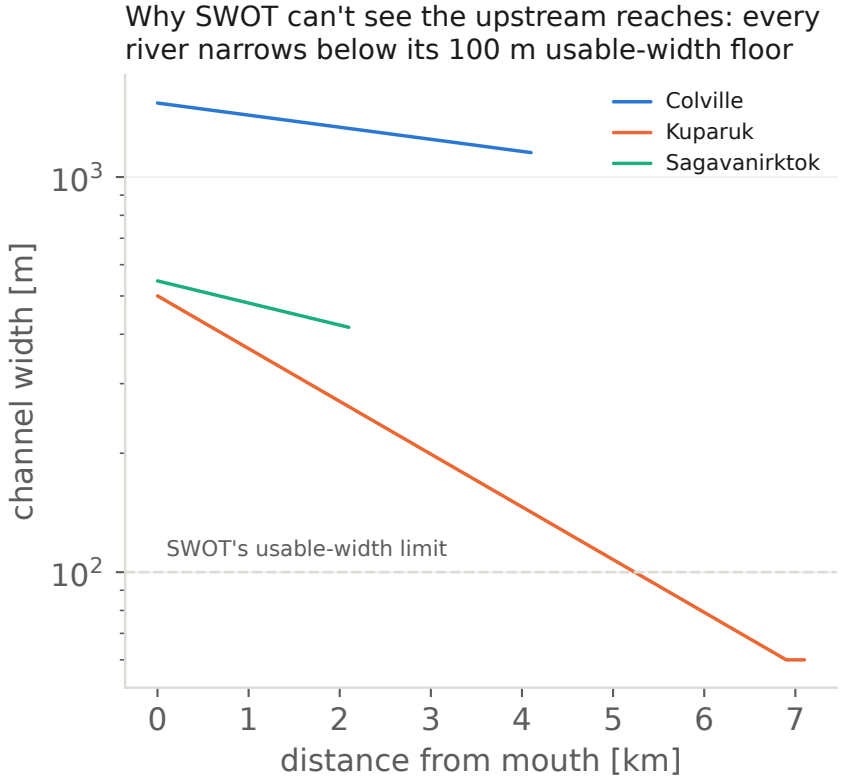


Colville: 6 example drifters released in Phase 2, each steadily carried out to sea by the river's current



Part 3 results: airborne imaging and the notional satellite channels





Part 4: What is a "Green's function," and why does this analysis need one?

The real goal of this exercise is to figure out which measurements would most improve confidence in the model's inputs -- both the BOUNDARY conditions (Part 1) and, as explained below, conditions INSIDE the channel.

A measurement inside the river doesn't directly tell you an input's value -- it tells you the RESULT of that input after it has traveled, mixed, and reacted its way through the estuary. To work backwards from "what did we measure" to "what does that imply," we need to know HOW SENSITIVE the model's interior state is to each input -- i.e., if the ocean's carbon content were 10% higher than assumed, how much would that actually change what a sensor sitting inside the river would read, at every place and time?

That sensitivity relationship is a GREEN'S FUNCTION, computed the most direct way possible: run the model once normally, run it again with ONE input turned up by 10%, and measure the difference.

The same idea extends naturally to unknowns that are NOT at either boundary: a local in-stream reaction rate, or a local source of organic matter entering ALONG the channel, can be turned up by 10% in one stretch of river at a time and the same recipe applied. This is new in this round of analysis -- 25 inputs are tested in total (next page): 19 at the two boundaries, plus 6 REACH-LOCAL ones inside the domain.

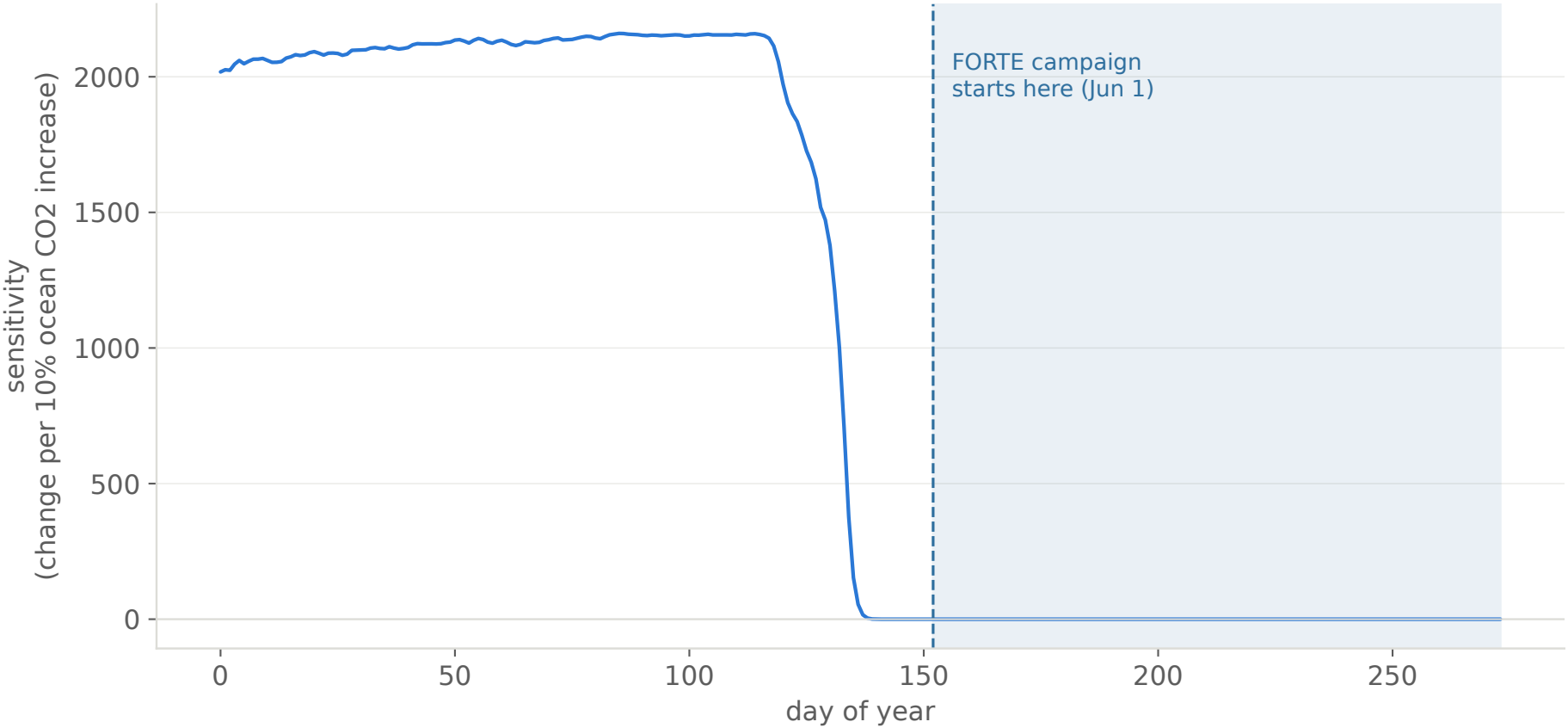
Part 4: what was actually tested

The 25 inputs tested	Which boundary	In plain terms
River discharge	DOWNSTREAM	How much water the river is carrying
Ocean temperature	DOWNSTREAM	Temperature of the seawater at the river mouth
Ocean salinity	DOWNSTREAM	Saltiness of the seawater at the river mouth
Ocean dissolved CO2 (DIC)	DOWNSTREAM	Carbon dioxide dissolved in the seawater
Ocean alkalinity	DOWNSTREAM	The seawater's capacity to buffer/absorb CO2
Ocean nitrate	DOWNSTREAM	A nutrient seaweed and algae need to grow
Ocean ammonium	DOWNSTREAM	Another nitrogen-based nutrient
Ocean phosphate	DOWNSTREAM	A nutrient needed for growth, often scarcest
Ocean dissolved silica	DOWNSTREAM	Needed by one type of algae (diatoms)
Ocean dissolved oxygen	DOWNSTREAM	How much oxygen is dissolved in the seawater
Ocean organic carbon	DOWNSTREAM	Dissolved organic matter in the seawater
River organic carbon	UPSTREAM	Dissolved organic matter coming down the river
River nitrate	UPSTREAM	A nutrient the river carries down from inland
River ammonium	UPSTREAM	Another nitrogen-based nutrient from inland
River phosphate	UPSTREAM	A nutrient from inland, often the scarcest
River dissolved silica	UPSTREAM	Needed by diatoms, carried down from inland
River dissolved CO2 (DIC)	UPSTREAM	Carbon dioxide the river carries from inland
River alkalinity	UPSTREAM	The river water's own buffering capacity
River algae (DIA)	UPSTREAM	Phytoplankton biomass entering from upstream
Degradation rate, mouth third	LOCAL (interior)	How fast organic carbon breaks down, near the mouth
Degradation rate, middle third	LOCAL (interior)	The same, in the middle stretch of the channel
Degradation rate, head third	LOCAL (interior)	The same, near the upstream/headwater end
Organic loading, mouth third	LOCAL (interior)	Extra organic carbon entering along the channel, near the mouth
Organic loading, middle third	LOCAL (interior)	The same, in the middle stretch of the channel
Organic loading, head third	LOCAL (interior)	The same, near the upstream/headwater end

Each was increased by 10% for an entire run (LOCAL ones, only within their own third of the channel). UPSTREAM rows (green) match what the campaign's grab sampling measures directly. LOCAL rows (blue) test whether spatially-distributed sampling can also resolve conditions INSIDE the channel, not just at its two ends -- see the along-estuary figures ahead.

Part 4, Finding 1 of 2: the ocean boundary goes dark

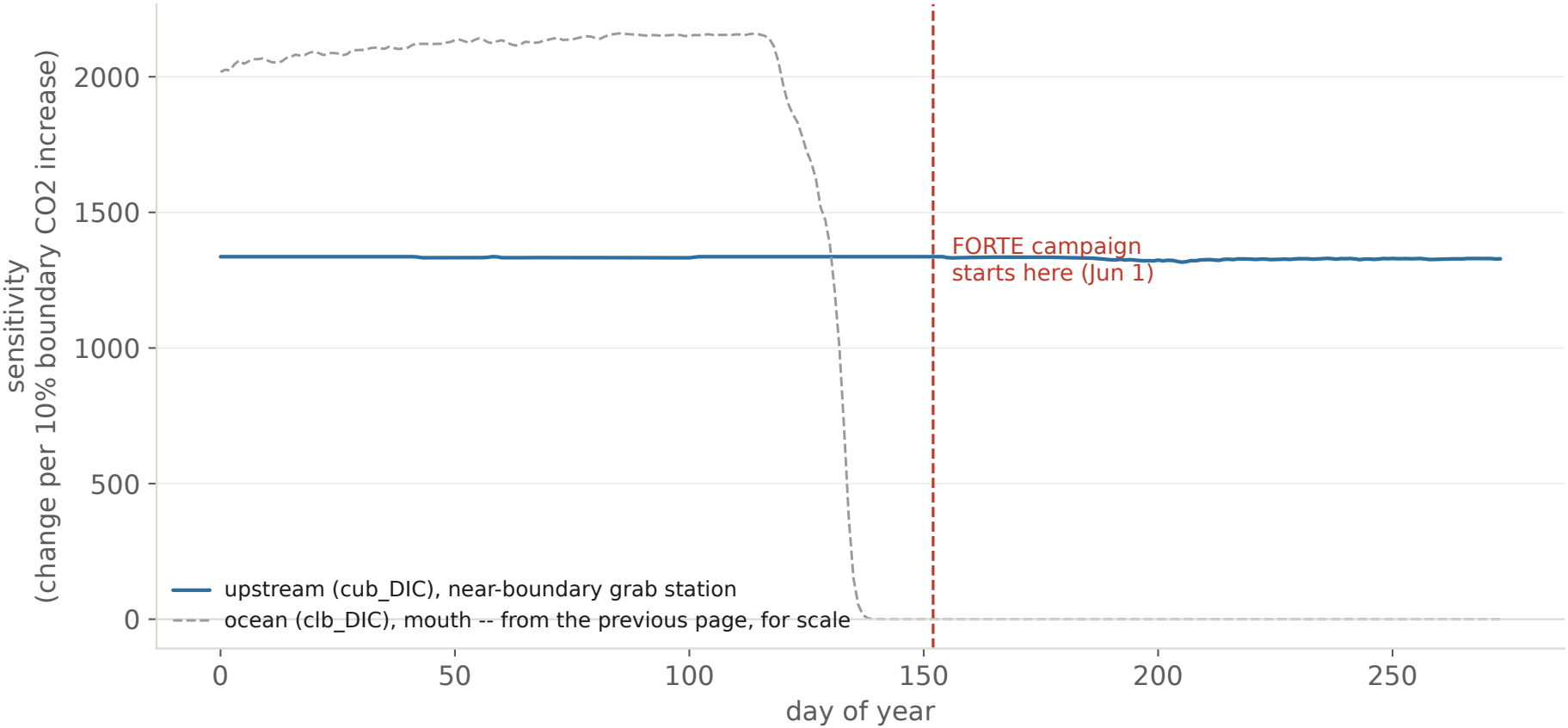
Colville: how much a 10% increase in the ocean's dissolved-CO2 level would change the river's own dissolved CO2, right at the mouth, by day of year.



The ocean's influence is strong all winter (the river is slow and salty, so ocean water sits inside it) and collapses to essentially ZERO right around day 140-150 -- just before the campaign begins. The spring flood has, by then, flushed the estuary completely fresh, cutting off any path for ocean water to reach anywhere the campaign samples. This is real: no instrument INSIDE the river can learn anything about the ocean boundary during FORTE's season. But it is only half the story -- see the next page.

Part 4, Finding 2 of 2: the RIVER boundary does not go dark

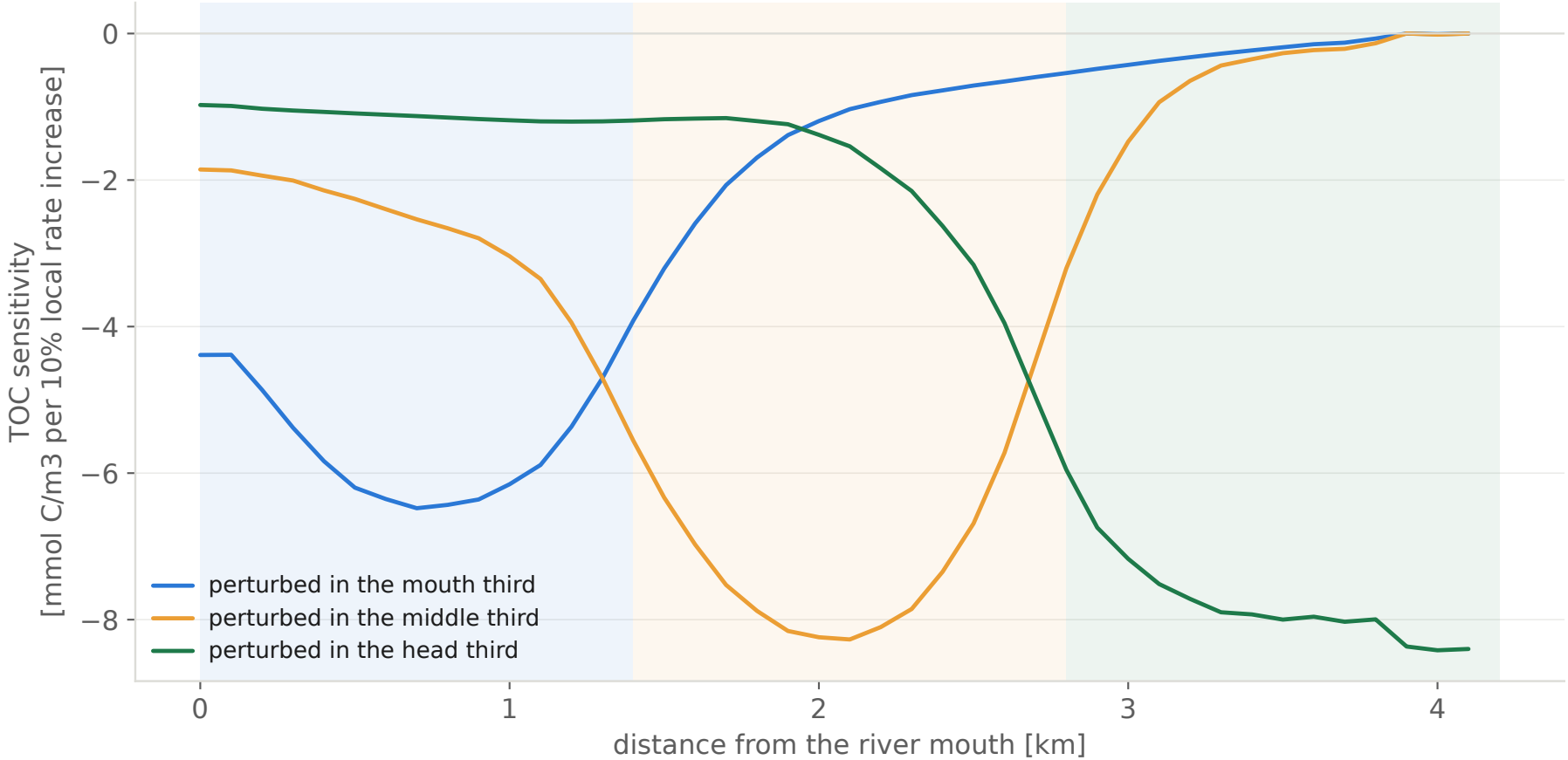
Colville: the SAME test, but now for a 10% increase in the RIVER's own upstream dissolved-CO2 level, measured at the grab-sampling station closest to the upstream boundary (3.9 of this river's 4.15 km).



Unlike the ocean signal, the upstream signal barely moves across the whole year -- it stays close to the boundary's own concentration throughout the FORTE window, because this station sits only 250 m from the boundary itself. This is why grab sampling -- despite learning nothing about the ocean boundary this season (Finding 1) -- still ends up the single most valuable channel overall (Part 5): it is directly measuring the river boundary.

Part 4: along-estuary sensitivity -- a local RATE perturbation stays local

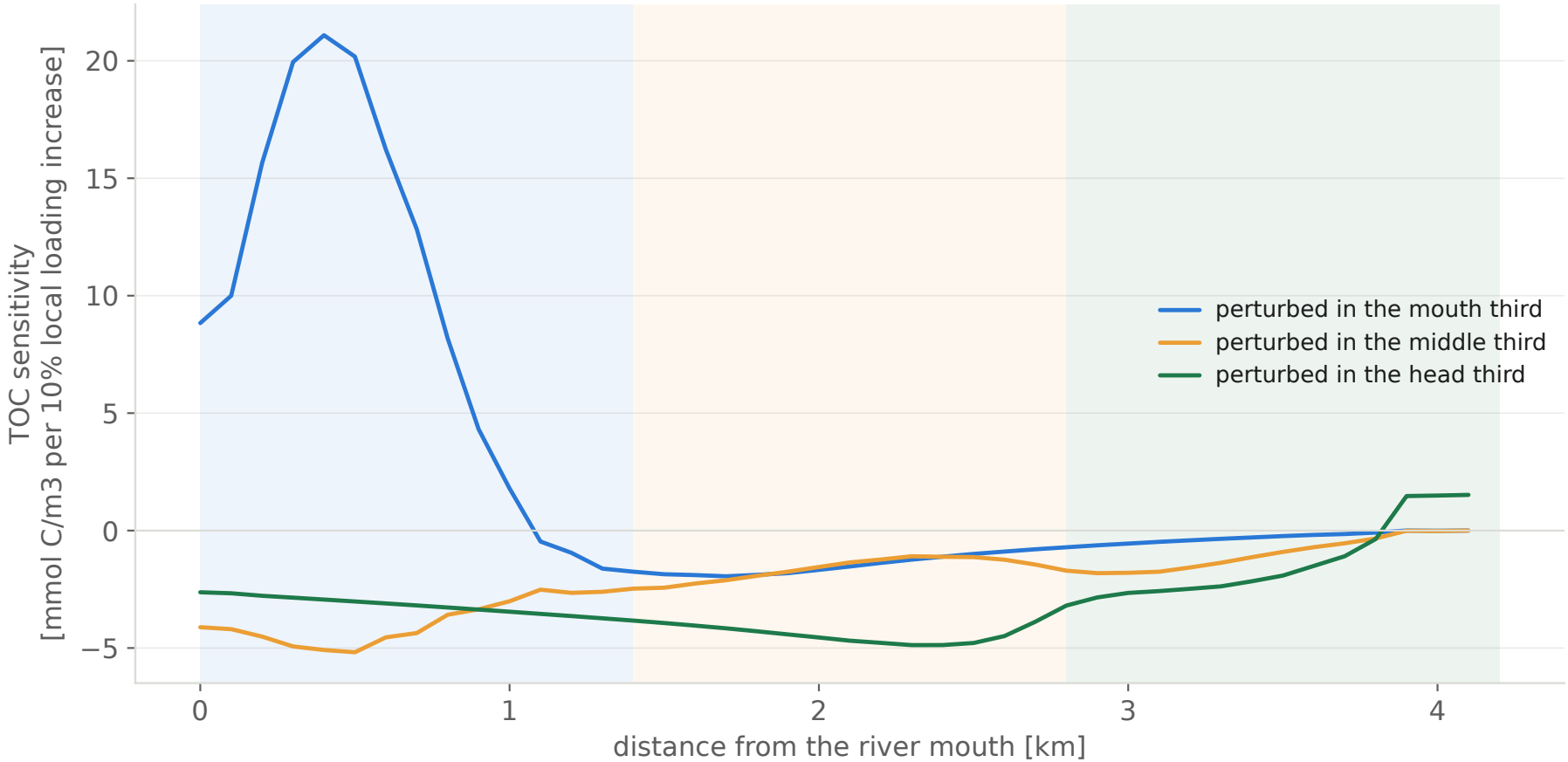
Colville: how much a 10% local increase in the aerobic-degradation rate -- applied in only ONE third of the channel at a time -- changes the river's own organic carbon (TOC) at every point along the channel (averaged over the whole campaign window, day-of-year 0-273).



Each curve is most intense (TOC drops the most, since a faster rate consumes more of it) inside its OWN perturbed third (tinted bands) and fades away from it -- a local rate change mostly shows up locally, with only a modest footprint carried into neighboring stretches by the river's own flow. This is what lets multi-station grab sampling (not a single fixed mooring) tell these three thirds apart (see the reach-by-reach ranking, page 30).

Part 4: along-estuary sensitivity -- a local LOADING perturbation, same test

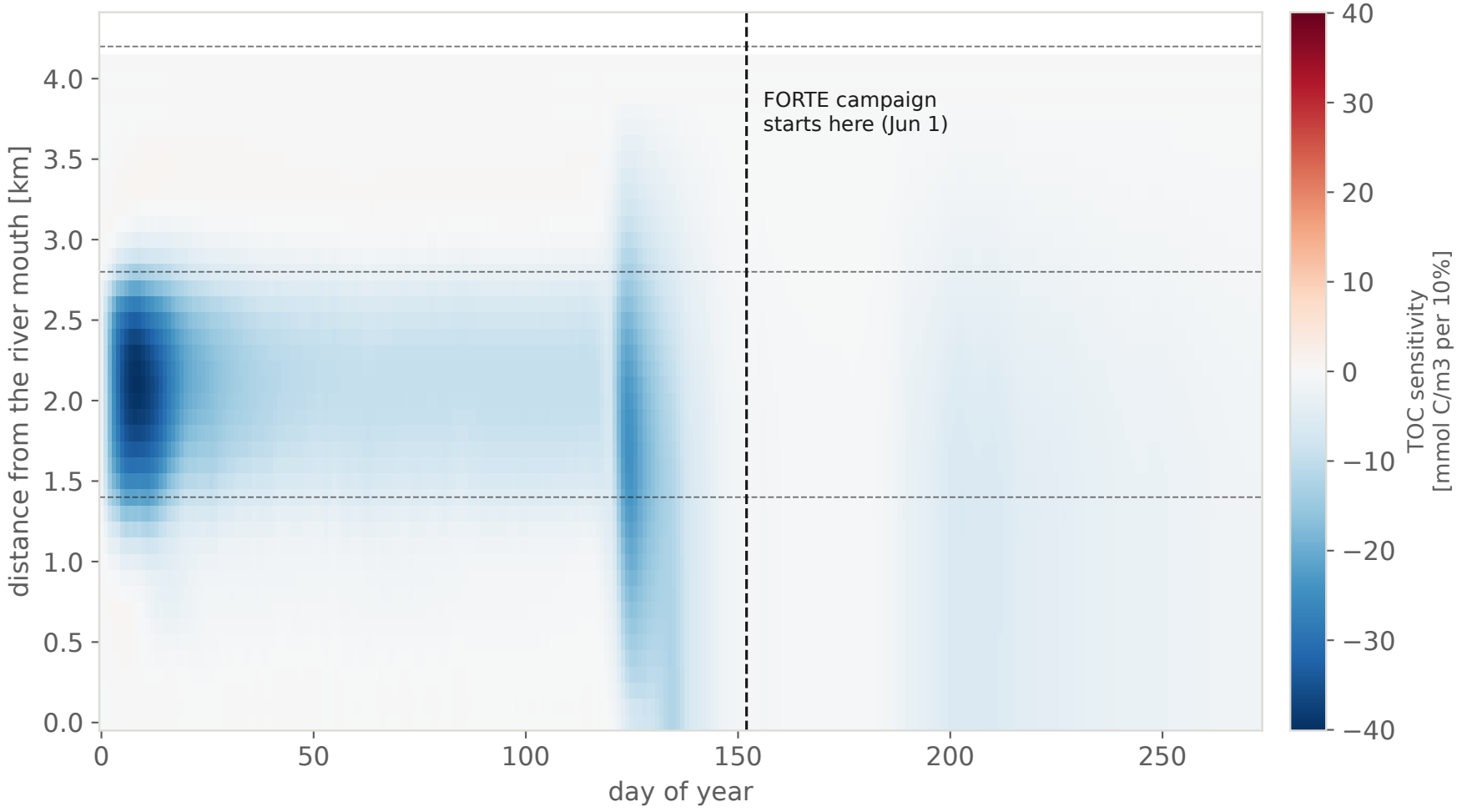
Colville: the SAME test, but now for a local 10% BOOST to organic-carbon entering along the channel (e.g. a tundra or thermokarst source), applied in only one third of the channel at a time.



Same localized-footprint pattern as the rate test -- and NOT a weaker signal (the mouth-reach peak is larger here; checked at the 5 real grab stations, the two are comparably strong, 0.81 vs 0.79 average). Loading still scores lower in the ranking ahead (page 30) because its pattern is more entangled with discharge (estimated at the same time), not because it is weaker.

Part 4: the full picture -- sensitivity across BOTH space and time

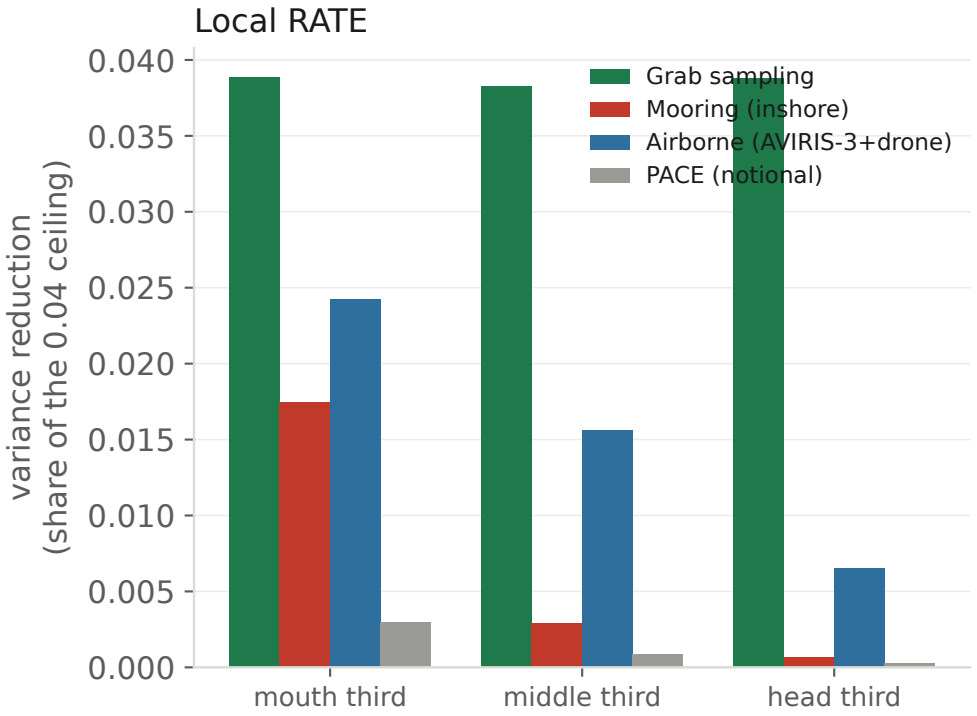
Colville: the same mid-third rate perturbation as the previous page, but now showing every day of the campaign season, not just the average -- a "Hovmoller" map, a standard way to show how a spatial pattern evolves over time in one picture.

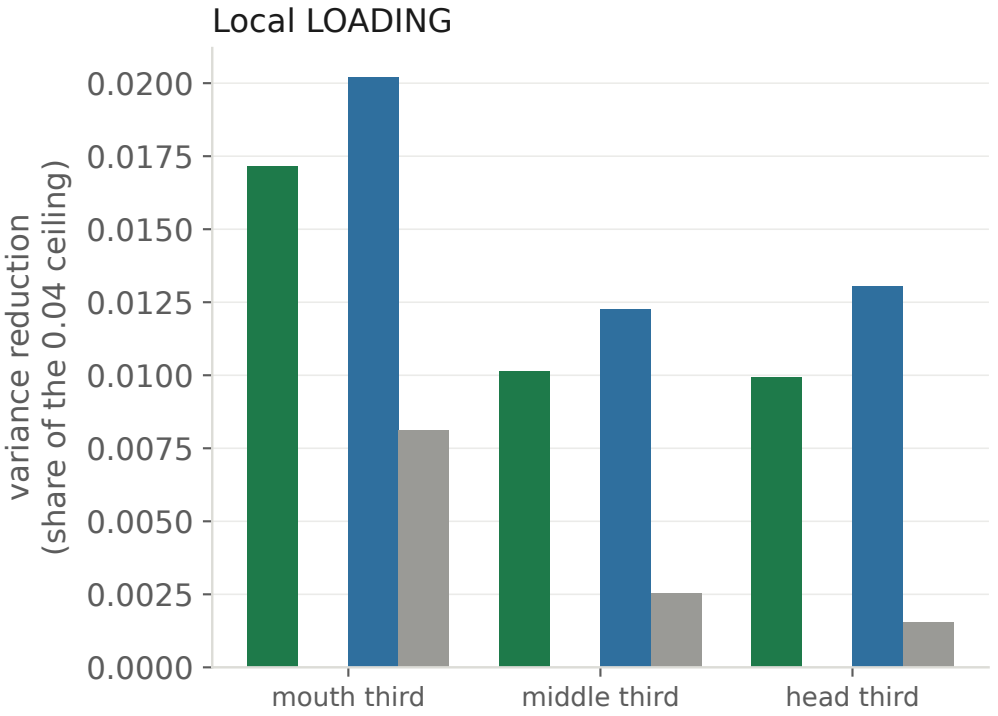


The bright band stays pinned to the perturbed middle third (dashed lines mark reach boundaries) MOST of the season -- it briefly spreads across the whole channel twice: early on (settling in), and around day 120-150, when the spring freshet mixes the whole channel (the same event behind Finding 1's collapse). Through the entire FORTE season, the local signal stays local.

Part 4: which channel resolves WHERE -- reach-by-reach value

The same information as the Part 5 ranking, split by reach instead of totaled -- this is what the along-estuary figures above predict should happen.





Grab (green) stays high in every reach -- its 5 stations span the whole estuary. The inshore mooring (red) is sharply LOCALIZED to its own fixed location (mouth) and nearly blind to the other two reaches, and contributes NOTHING to local loading at all -- it does not measure organic carbon.

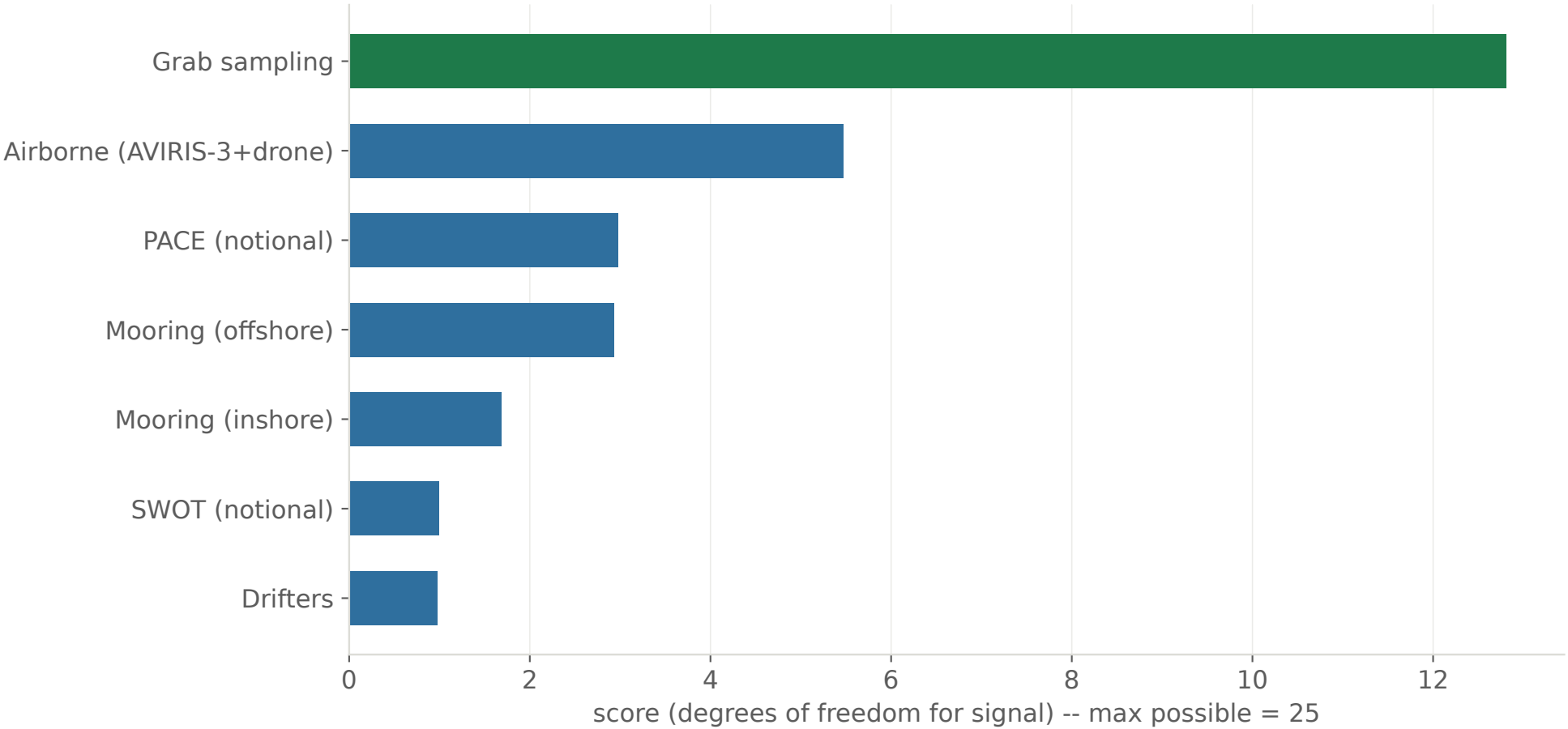
Part 5: How "which measurement is most valuable" is actually scored

For each synthetic measurement (Part 3), combined with how sensitive the model is to each input at that place and time (Part 4), and how precise the instrument actually is, this analysis computes how much that single measurement would have narrowed down our uncertainty about each of the 25 inputs tested. Adding this up correctly across every measurement a channel takes gives a single score per channel, called DEGREES OF FREEDOM FOR SIGNAL (DFS) -- a standard information-theory score used throughout atmospheric and ocean data assimilation.

A score of 0 means a channel taught us nothing; a score of 25 (the number of inputs tested) would mean it fully pinned down every single one. In practice, scores land somewhere in between, and the split BETWEEN inputs (which ones got constrained, which didn't) matters as much as the total -- a channel can score well by fully nailing down a FEW inputs (as grab sampling does, for the upstream and along-estuary ones) rather than partially touching all 25.

ONE IMPORTANT CORRECTION APPLIED: a sensor recording every 6 minutes for months doesn't really deliver thousands of independent pieces of evidence -- consecutive readings are almost identical. This analysis explicitly corrects for that (an "effective sample size" adjustment), so a flood of repetitive readings is not mistaken for a flood of new information.

Part 5: final ranking -- which measurement actually taught us the most



Grab sampling (highlighted) comes out on top -- not from the ocean boundary (Part 4, Finding 1: cut off all season) but from the RIVER's own upstream boundary (Finding 2), PLUS a further boost from along-estuary sensitivity. Airborne and PACE pick up smaller versions of both effects.

Part 5: three different effects, disentangled -- timing, upstream, along-estuary

Measurement type	(A) Full year, ocean+discharge only (earlier analysis)	(B) FORTE window, ocean+discharge only (isolates TIMING alone)	(C) FORTE window, + upstream controls (isolates UPSTREAM alone)	(D) FORTE window, + along-estuary controls (this report's full result)
Grab sampling	5.62 / 11	1.00 / 11	8.99 / 19	12.81 / 25
Mooring (inshore)	2.62 / 11	0.98 / 11	1.32 / 19	1.69 / 25
Mooring (offshore)	2.99 / 11	2.94 / 11	2.94 / 19	2.94 / 25
Drifters	1.93 / 11	0.97 / 11	0.97 / 19	0.97 / 25
Airborne (AVIRIS-3+drone)	1.61 / 11	1.00 / 11	3.20 / 19	5.47 / 25
SWOT (notional)	1.00 / 11	1.00 / 11	1.00 / 19	1.00 / 25
PACE (notional)	2.68 / 11	0.75 / 11	2.60 / 19	2.97 / 25

Reading grab sampling's row left to right: (A -> B) TIMING ALONE crashes its value 5.62 -> 1.00 -- the summer window cannot see the ocean boundary (Finding 1). (B -> C) The 8 upstream controls restore it to 8.99/19 -- grab was ALWAYS also measuring the river boundary, column (B) just never asked (Finding 2). (C -> D) The 6 along-estuary controls lift it further to 12.81/25 -- grab's 5 stations also resolve WHERE along the channel things happen, a question boundary-only testing cannot even pose. Airborne shows the same 3-step pattern; moorings/drifters/SWOT barely move in C or D, since they are not well-positioned to sense either structure.

Part 6: Conclusions and practical implications

1. THE LONG-TERM RECORD (Part 1) shows a significant, coherent dilution signal at all three rivers: discharge and DOC export trending up, DIC/alkalinity trending down. Freshet timing, ice duration, and most reaction rates show no significant trend over this 19-year baseline -- reported as a genuine null result where that is what the data shows.

2. WITHIN the FORTE campaign window (Jun 1 - Sep 30), no instrument -- including grab sampling -- can learn anything about the OCEAN boundary. The spring flood has already cut that connection by the time the campaign starts. The ONE exception is a mooring placed directly in the ocean, which reads that boundary directly.

3. BUT grab sampling is still the single MOST valuable channel overall -- it also, independently, measures the RIVER's own upstream boundary (its headwater stations read it almost directly), AND resolves WHERE along the channel local changes in rate or loading are happening, something no boundary-only test could even pose as a question.

4. A single fixed mooring, by contrast, is sharply localized to its own location and contributes nothing to resolving local organic-carbon loading anywhere -- it is a fundamentally different, complementary measurement, strongest for the ocean boundary specifically.

5. Practical reading for campaign planning: grab sampling's real strength, as currently scheduled, is characterizing what the RIVER is carrying and WHERE along its length -- not the ocean. If the ocean boundary matters too, extend monitoring into the cold season or rely on the offshore mooring, which is immune to the seasonal blind spot.

CAVEATS: results come from a model, calibrated against real rivers but not identical to them; the sensitivity analysis assumes one "typical" campaign season behaves like every year; a single, equal 20% uncertainty was assumed for every input, since no better estimate exists for most of them; and 7 of 8 upstream controls plus all 6 along-estuary controls are tested as simple step changes, not genuine time-varying or continuously-distributed inputs. These affect exact numbers, not the findings above, which rest on simple, robust mechanisms not sensitive to any of these choices.